

The empirical recurrence for OEIS A264269, recovered from its tail

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Abstract

OEIS A264269 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 97$ — every index at which it can be written down. Its last coefficient is $c_{96} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A264269 is “Number of $(n+1)X(5+1)$ arrays of permutations of $0..n*6+5$ with each element having index change $+(-,.)$ 0,0 1,0 or 1,2.”. It has offset 1 and begins

169, 6400, 443556, 25806400, 1608491236, 98054154496, 6027088830169, 369450492999025, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Nov 10 2015 11:31:16, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	62	c_{25}	−230478295837205752238090	c_{49}	12331883755839031994311352953
c_2	970	c_{26}	−247730798247991140088286	c_{50}	−3722628457501823209738693619
c_3	−64534	c_{27}	2340875461147626403466482	c_{51}	−9089280250614674470025887961
c_4	−236042	c_{28}	122139561086756523706174	c_{52}	5268297155646449258422411507
c_5	25782014	c_{29}	−17502081739977690500683594	c_{53}	4795772298128815877563506866
c_6	−25547618	c_{30}	12345579457913033996447222	c_{54}	−4426680344458647021885919139
c_7	−5147533362	c_{31}	96428215577437869291640838	c_{55}	−1637437764035537715445404187
c_8	19072160398	c_{32}	−133211617909078211247851370	c_{56}	2604006732923221934916797505
c_9	576009618186	c_{33}	−383503326952991720365062894	c_{57}	2007221170487354560688896990
c_{10}	−3300465784158	c_{34}	828489351174488431952368570	c_{58}	−1121517646117252479677539781
c_{11}	−38569431920754	c_{35}	1027274600284360866074479046	c_{59}	1370874030829010289433881642
c_{12}	301173784241286	c_{36}	−3570029766474115157775698290	c_{60}	3570029766474115157775698290
c_{13}	1568001501715454	c_{37}	−1370874030829010289433881642	c_{61}	−102727460028436086607447904
c_{14}	−17174448835191006	c_{38}	11215176461172524796775397818	c_{62}	−828489351174488431952368570
c_{15}	−34867055939148422	c_{39}	−2007221170487354560688896990	c_{63}	383503326952991720365062894
c_{16}	660275195419827502	c_{40}	−26040067329232219349167975050	c_{64}	133211617909078211247851370
c_{17}	90122668042459454	c_{41}	16374377640355377154454041878	c_{65}	−96428215577437869291640838
c_{18}	−17805280929354316114	c_{42}	44266803444586470218859191398	c_{66}	−12345579457913033996447222
c_{19}	21199400907247922478	c_{43}	−47957722981288158775635068666	c_{67}	17502081739977690500683594
c_{20}	342863906809634610742	c_{44}	−52682971556464492584224115074	c_{68}	−122139561086756523706174
c_{21}	−789350032017008023078	c_{45}	90892802506146744700258879618	c_{69}	−2340875461147626403466482
c_{22}	−4702841749669939507990	c_{46}	37226284575018232097386936194	c_{70}	247730798247991140088286
c_{23}	16359485540104235900782	c_{47}	−123318837558390319943113529530	c_{71}	230478295837205752238090
c_{24}	44303867695470804291058	c_{48}	0	c_{72}	−44303867695470804291058

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A264269 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{95} - \dots - c_{96}$. Its constant term is $-c_{96} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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